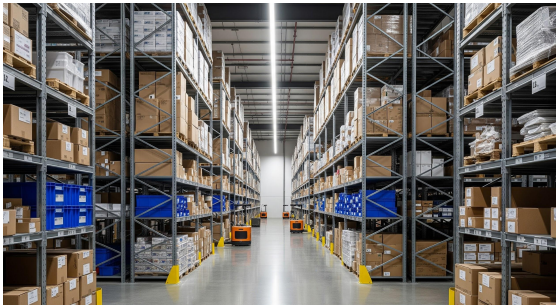
MGT 207
Inventory Management

Cycle, Pipeline, EOQ, Safety Stock, Newsvendor & (Q, r) Models

What is Inventory?

Inventory: A stock of materials used to satisfy customer demand or support production.



Key Purposes

- **Customer Demand:** Ready to fulfill orders immediately without delay.
- **Production Support:** Supplies and components for manufacturing goods or providing services.
- **Operational Buffer:** Managing variability and decoupling stages across supply chains.

Pressures for Small Inventories

Inventory Holding Cost

The primary pressure to keep inventory low is the significant expense of maintaining stock over time.

Cost Components

- **Cost of Capital:** Opportunity cost of money tied up in stock.
- **Storage & Handling:** Warehouse rent, utilities, labor, and equipment.
- **Taxes & Insurance:** Property taxes and insurance coverage.
- **Inventory Shrinkage:** Pilferage (theft), obsolescence, and deterioration/spoilage.

Pressures for Large Inventories

Drivers for Higher Stock

Operational and financial pressures encouraging larger inventory levels:

Key Pressures

- **Customer Service & Stockouts:** Prevents lost sales and maintains customer goodwill.
- **Ordering & Setup Costs:** Spreads fixed costs over more units.
- **Labor & Equipment Efficiency:** Reduces setup frequency and disruptions.
- **Transportation Economies:** Full truckload (FTL) discounts.
- **Payments to Suppliers:** Quantity discounts on purchase price.

Types of Inventory

Operational Inventories

Classified by the primary operational purpose they serve in the supply chain:

Cycle Inventory

Inventory that results directly from the **lot size** or batch size used in production or purchasing.

Driven by fixed ordering/setup costs.

Safety Stock

Buffer stock held to protect against **uncertainties** in customer demand, lead time, or supplier delivery.

Prevents costly stockouts.

Pipeline Inventory

Inventory that is **in transit** between different locations (plants, warehouses, stores) in the supply chain.

Driven by transit lead time.

Cycle Inventory

Lot Sizing Principles

- The lot size, Q, varies directly with the elapsed time (or cycle) between orders.
- The longer the time between orders for a given item, the greater the cycle inventory must be.

Cycle Inventory Calculation

$$\textbf{Average cycle inventory} = \frac{Q + 0}{2} = \frac{Q}{2}$$

Pipeline Inventory

Components and Definitions

Lead Time Demand	Periodic Demand	Lead Time Duration
(Average) demand during lead time:	(Average) demand for items per period:	Number of periods in item's lead time:
D_L	$\bar{d}$	L

Pipeline Inventory Calculation

$$\textbf{Pipeline inventory} = D_L = \bar{d} \times L$$

Example 1: Electric Drills (1 of 3)

Scenario Details

A plant makes monthly shipments of electric drills to a wholesaler.

The wholesaler pays for inventory from the moment the plant makes a shipment.

The Challenge: What is the effect on the wholesaler's cycle and pipeline inventories?

Current Status

- **Lot Size (Q):** 280 drills
- **Demand ($\bar{d}$):** 70 drills / week
- **Lead Time (L):** 3 weeks

New Proposal

- **Lot Size (Q):** 350 units
- **Demand ($\bar{d}$):** 70 drills / week
- **Lead Time (L):** 2 weeks (Guaranteed)

Example 1: Current Policy (2 of 3)

The wholesaler's current cycle and pipeline inventories are calculated as follows:

Cycle Inventory

Formula:

$$\text{Cycle Inventory} = \frac{Q}{2}$$

Calculation:

$$\frac{280}{2} = \textbf{140 drills}$$

Pipeline Inventory

Formula:

$$\text{Pipeline Inventory} = D_L = \bar{d} \times L$$

Calculation:

$$(70 \text{ drills/week}) \times (3 \text{ weeks}) = \textbf{210 drills}$$

Example 1: Proposed Policy (3 of 3)

The wholesaler's cycle and pipeline inventories under the new proposal:

Cycle Inventory

Formula:

$$\text{Cycle Inventory} = \frac{Q}{2}$$

Calculation:

$$\frac{350}{2} = \textbf{175 drills}$$

Pipeline Inventory

Formula:

$$\text{Pipeline Inventory} = D_L = \bar{d} \times L$$

Calculation:

$$(70 \text{ drills/week}) \times (2 \text{ weeks}) = \textbf{140 drills}$$

Trade-off Insight

Larger lot size increases cycle inventory ($+35$ drills), but faster delivery reduces pipeline inventory (-70 drills), netting an overall inventory reduction of 35 drills!

Safety Stock Inventory

Definition

Extra inventory held to protect against **demand** or **supply** uncertainty.

(We will calculate safety stock explicitly in the probabilistic EOQ and (Q, r) models later.)

Next Step

Now that we understand the three main inventory types, let's look at the costs that drive decisions about them.

Inventory Costs: The Two Big Ones

Holding (Carrying) Cost (H)

Cost of keeping one unit in inventory for one year (capital, storage, handling, insurance, obsolescence).

Ordering / Setup Cost (S)

Fixed cost every time we place an order or set up a production run (independent of order size Q).

Fundamental Tension

These two costs are in direct conflict — this fundamental tension creates the need for the **EOQ model**.

Economic Order Quantity (1 of 2)

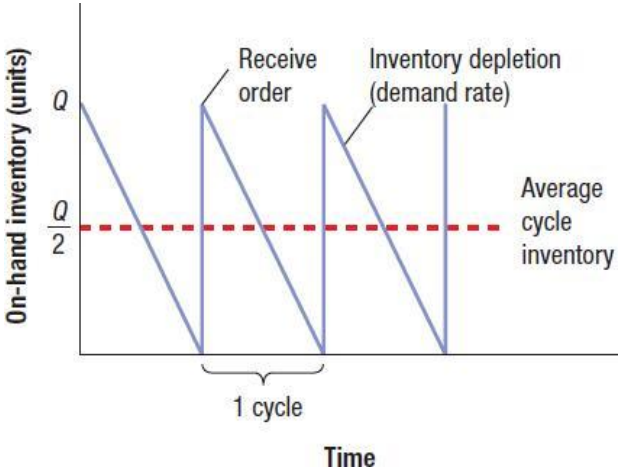
Economic Order Quantity (EOQ): The lot size, Q, that minimizes total annual inventory holding and ordering costs.

Five Key Assumptions

1. The demand rate is constant and known with certainty (will be relaxed).
2. No constraints are placed on the size of each lot (no truckload or capacity limits).
3. The only two relevant costs are inventory holding cost (H) and fixed ordering/setup cost (S).
4. Decisions for one item can be made independently of decisions for other items.
5. The lead time is constant and known with certainty (will be relaxed).

Calculating EOQ: Cycle-Inventory Levels

Figure 9.5: Cycle-Inventory Levels (The Sawtooth Curve)



Calculating EOQ: Cost Equations

Annual Holding Cost

$$\text{Annual holding cost} = (\text{Average cycle inventory}) \times (\text{Unit holding cost}) = \frac{\mathbf{Q}}{\mathbf{2}} \cdot \mathbf{H}$$

Annual Ordering Cost

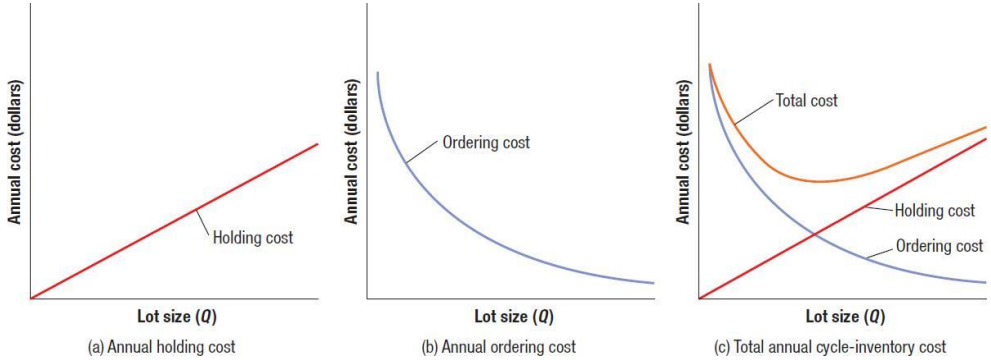
$$\text{Annual ordering cost} = (\text{Number of orders/Year}) \times (\text{Ordering/setup cost}) = \frac{\mathbf{D}}{\mathbf{Q}} \cdot \mathbf{S}$$

Total Annual Cost (C)

$$\text{Total cost} = \text{Annual holding cost} + \text{Annual ordering cost} = \frac{\mathbf{Q}}{\mathbf{2}}(\mathbf{H}) + \frac{\mathbf{D}}{\mathbf{Q}}(\mathbf{S})$$

Calculating Total Inventory Cost

Figure 9.6: Graphs of Annual Holding, Ordering, and Total Costs



Total Inventory Cost

Total Annual Cycle-Inventory Cost

$$C = \frac{Q}{2}(H) + \frac{D}{Q}(S)$$

Variable Definitions

- C = total annual cycle-inventory cost
- Q = lot size (in units)
- H = holding cost per unit per year
- D = annual demand (in units per year)
- S = ordering or setup costs (in dollars per lot)

Example 2: Museum Gift Shop Bird Feeder (1 of 3)

Problem Context

- Museum gift shop operates **52 weeks/year**.
- Top-selling SKU: **Bird feeder**.
- Current policy: **468-unit lot size**.

Cost Parameters

- Sales (d): **18 units / week**
- Supplier Charge (P): **\$60 / unit**
- Ordering Cost (S): **\$45 / order**
- Holding Cost (H): **25% of feeder's value**

Key Questions to Solve:

1. What is the annual cycle-inventory cost of the current 468-unit lot size policy?
2. Would a lot size of **390** be better?

Example 2: Comparing Lot Sizes (2 of 3)

Step 1: Compute Annual Demand (D) and Holding Cost (H)

$$D = (18 \text{ units/week}) \times (52 \text{ weeks/year}) = \mathbf{936 \text{ units}}$$

$$H = 0.25 \times (\$60/\text{unit}) = \mathbf{\$15}$$

Alternative Lot Size ($Q = 390$)

$$C = \frac{390}{2}(\$15) + \frac{936}{390}(\$45)$$

$$C = \$2,925 + \$108 = \mathbf{\$3,033}$$

Current Policy ($Q = 468$)

$$C = \frac{468}{2}(\$15) + \frac{936}{468}(\$45)$$

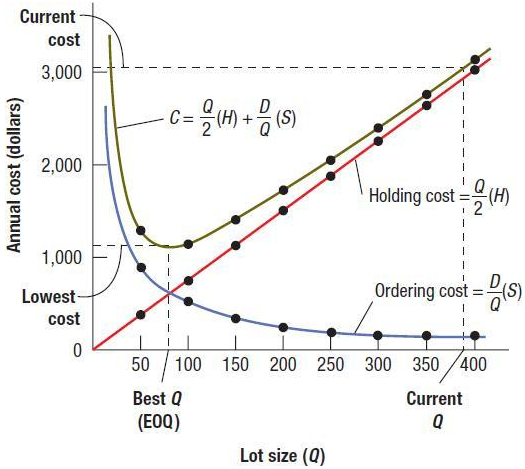
$$C = \$3,510 + \$90 = \mathbf{\$3,600}$$

Result

The alternative lot size ($Q = 390$) results in a lower annual cost of **\$3,033** compared to **\$3,600** (savings of \$567/year).

Example 2: Cost Function Curve (3 of 3)

Total Annual Cycle-Inventory Cost Function for the Bird Feeder:



Key Insight

Min total cost is reached exactly when holding cost and ordering cost cross each other!

Deriving the EOQ Formula

Objective: Minimize Total Annual Cost (TC)

The Derivation Steps:

- Set Holding Cost equal to Ordering Cost:
  $$\frac{Q}{2}H = \frac{D}{Q}S$$
- Multiply both sides by Q to consolidate:
  $$\frac{Q^2}{2}H = DS$$
- Isolate Q^2 by multiplying by 2 and dividing by H:
  $$Q^2 = \frac{2DS}{H}$$
- Take the square root to find optimal Q:
  $$Q = \sqrt{\frac{2DS}{H}}$$

Economic Order Quantity

$$Q = \sqrt{\frac{2DS}{H}}$$

Q: Optimal order quantity
D: Annual demand in units
S: Order cost per purchase order
H: Annual holding cost per unit

Note: Assumes total cost curve is convex where first derivative equals zero: $\frac{dC}{dQ} = \frac{H}{2} - \frac{DS}{Q^2} = 0$.

EOQ Review So Far...

1. Economic Order Quantity (EOQ)

$$\text{EOQ} = \sqrt{\frac{2DS}{H}}$$

2. Time Between Orders (TBO)

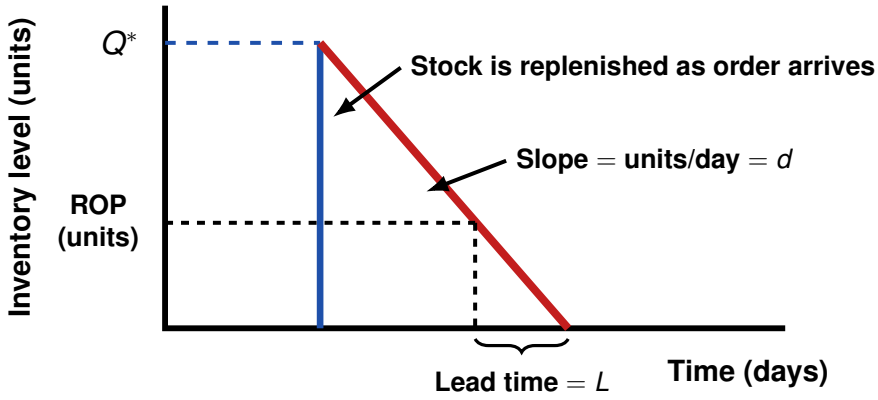
$$\text{TBO}_{\text{EOQ}} = \frac{\text{EOQ}}{D} \times (12 \text{ months / year})$$

The Missing Piece

When do you order?

What is missing to implement inventory management using EOQ?

Reorder Point Curve



Reorder Points

- **The reorder point (ROP)** tells *"when"* to order.
- **Lead time (*L*)** is the time between placing and receiving an order.

$$\textbf{ROP} = (\textbf{Demand per day}) \times (\textbf{Lead time in days})$$
$$\textbf{ROP} = d \times L$$

Daily Demand Rate (d)

$$d = \frac{D}{\text{Number of working days in a year}}$$

Reorder Point Example

Parameters

- Demand = **8,000 units per year**
- **250 working days per year**
- Lead time for orders is **3 working days**, may take **4 days**

Step 1: Compute Daily Demand (d)

$$d = \frac{D}{\text{Number of working days}} = \frac{8,000}{250} = \textbf{32 units/day}$$

Step 2: Compute Reorder Point (ROP $= d \times L$)

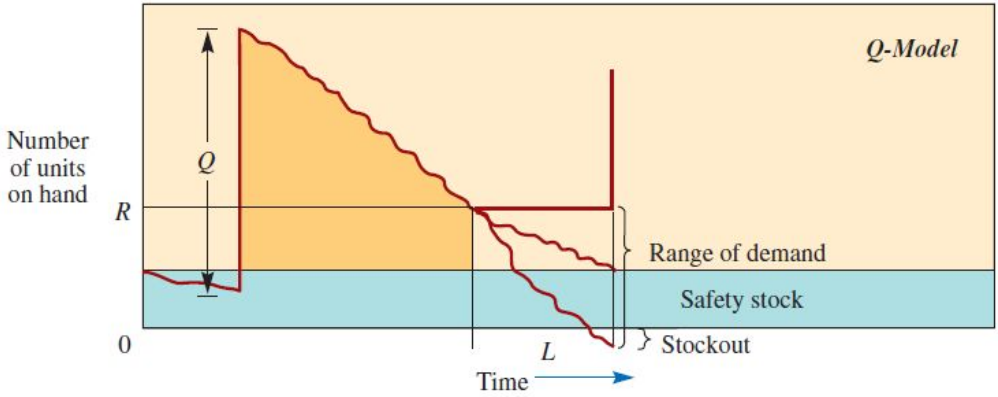
- For $L = 3$ days: ROP $= 32$ units/day $\times$ 3 days $=$ **96 units**
- For $L = 4$ days: ROP $= 32$ units/day $\times$ 4 days $=$ **128 units**

Probabilistic Fixed-Order Quantity (EOQ) Model

- In the fixed order quantity model, the ordering process is triggered when the inventory level drops to a critical point, the **Reorder Point (R)**.
- This starts the **lead time** for the item.
- During the lead time, customers continue to draw down the inventory.
- **If demand is uncertain, during this period the inventory is vulnerable to stockout** (run out of inventory).

Inventory On Hand: The Stochastic Q-Model

Uncertain Demand During Lead Time and Safety Stock Protection



Probabilistic Fixed-Order Quantity (EOQ) Model

- The customer **service level** is converted into a z value using the standard normal distribution table.

Safety Stock (SS)

$$\mathbf{SS} = \mathbf{z} \cdot \sigma_{\mathbf{L}} = \mathbf{z}\sqrt{\mathbf{L}}\sigma_{\mathbf{d}}$$

Expected Demand During Lead Time (EDDLT)

$$\mathbf{EDDLT} = \bar{\mathbf{d}} \cdot \mathbf{L}$$

Reorder Point (R)

$$\mathbf{R} = \mathbf{EDDLT} + \mathbf{SS} = \bar{\mathbf{d}} \cdot \mathbf{L} + \mathbf{z}\sqrt{\mathbf{L}(\sigma_{\mathbf{d}})^{\mathbf{2}}}$$

Sell-Rite Discount Stores: Problem Setup

Toy Inventory at Sell-Rite Discount Stores

Data indicates a **constant lead time (L) of 10 days** and a **normally distributed demand** with a mean of **1,250 toys/day** and a standard deviation of **375 toys/day**.

Questions to Solve:

a. **What is the EDDLT for the toy?** (Expected Demand During Lead Time)

b. **What is the standard deviation of DDLT for the toy?**

c. **Compute the reorder point (ROP) for the toy** (Service level specified at 90% during lead time).

d. **What is the safety stock?** (The buffer inventory held to prevent stockouts).

Sell-Rite Discount Stores: Solution

Given Parameters

$L = 10$ days, $\bar{d} = 1,250$ toys/day, $\sigma_d = 375$ toys/day

a. **EDDLT:** $\bar{d}L = 10(1,250) = \mathbf{12,500}$ **toys**

b. **Standard Deviation of Lead Time Demand:**

$$\sigma_L = \sqrt{L\sigma_d^2} = 375\sqrt{10} = \mathbf{1,186}\ \textbf{toys}$$

c. **Reorder Point (R):** From the Normal table, a 90% service level corresponds to $\mathbf{z = 1.28}$:

$$R = EDDLT + z\sigma_L = 12,500 + 1.28(1,186) = \mathbf{14,018}\ \textbf{toys}$$

d. **Safety Stock (SS):**

$$SS = z\sigma_L = 1.28(1,186) = \mathbf{1,518}\ \textbf{toys}$$

Transitioning: From EOQ to Newsvendor

Moving from deterministic cost minimization to betting on uncertain demand

EOQ Model: Cost-Based Logic

Focuses on optimal balance between **holding** and **ordering** costs.

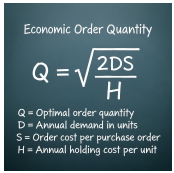
- Constant, known demand
- Deterministic lead times
- Goal: Minimize total annual cost

Newsvendor: Inventory Betting

A "one-shot" decision under **uncertain demand**.

- Stochastic, random demand
- Perishable / short-lifecycle goods
- Goal: Maximize expected profit

The Trade-off:
Overage vs. Underage Cost



The Risk:
Waste vs. Lost Sales

Inventory Model for Perishable Goods

Betting on Demand

The Classic Newsvendor Problem

Betting on Demand

Consider the problem of deciding how many loaves of bread to bake every day

Under-stocking Risk (Too Few)

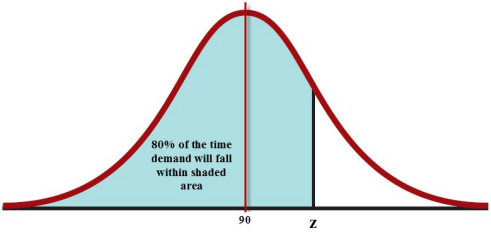
Too few loaves and some customers will not be able to purchase bread; profits associated with these potential sales are permanently lost.

Over-stocking Risk (Too Many)

Too many loaves lead to waste and unsold inventory that must be scrapped or sold at a steep discount at the end of the day.

How Many Loaves for an 80% Service Level?

- Suppose demand follows a **Normal distribution** with mean $\mu = 90$ loaves and standard deviation $\sigma = 10$ loaves.



- Stocking 90 loaves (average demand) yields a **50% service level** (chance of not stocking out).
- We want an **80% service level** (chance of not stocking out), i.e., $P = 80\%$.

Solving the Stocking Problem (80% Target)

$G(z)$	z
0.50	0
0.55	0.126
0.60	0.253
0.65	0.385
0.70	0.524
0.75	0.674
0.80	**0.842**
0.85	1.036

Excel: `"=NORMSINV(0.8)"` $\cong 0.842$

Calculation

- From z-table, **0.842 standard deviation** of "extra loaves" are needed away from mean demand 90.
- Standard deviation is $\sigma = 10$.
- Number of extra loaves:

$$0.842 \times 10 \cong \textbf{8 loaves}$$

- Stocking quantity:

$$90 + 8 = \textbf{98 loaves}$$

Pricing Information

Cost and Price Parameters

Suppose that:

- Each loaf sells for **$4.50** (Selling Price, p)
- The baker incurs **$1.50** in unit cost (Cost, c)
- Day-old bread can be sold for **$0.50** per loaf (Salvage, s)

Critical Question

Is the optimal stocking level still $90 + 8 = 98$ loaves?

Optimal Service Level P

There is an optimal way of choosing service level P based on economics:

The Critical Fractile Formula

$$P = \frac{C_u}{C_o + C_u}$$

Cost Definitions

- $\mathbf{C_o} = (\mathbf{c} - \mathbf{s})$: Unit overage cost (loss incurred for each unsold unit)
- $\mathbf{C_u} = (\mathbf{p} - \mathbf{c})$: Unit underage cost (profit lost for each unit of unmet demand)

In Our Case: Computing Costs

Calculating Overage and Underage Costs

- $C_u = \$4.50 - \$1.50 = \$3.00$ (Underage cost)
- $C_o = \$1.50 - \$0.50 = \$1.00$ (Overage cost)
- $C_u + C_o = \$3.00 + \$1.00 = \$4.00$

Optimal Service Level Result

Critical Ratio Calculation

$$C_u = \$3, \quad C_o = \$1$$

$$P = \frac{C_u}{C_o + C_u} = \frac{3}{1+3} = \frac{3}{4} = \mathbf{75\%}$$

Economic Takeaway

So, 80% was not optimal, 75% is!

We would incur a lower expected total cost with a **75% service level** than with an 80% service level.

Solving the Stocking Problem (75% Target)

$G(z)$	z
0.50	0
0.55	0.126
0.60	0.253
0.65	0.385
0.70	0.524
0.75	**0.674**
0.80	0.842
0.85	1.036

Optimal Stocking Calculation

- From z-table, **0.674 standard deviation** of extra loaves are needed away from mean demand 90.
- Standard deviation is $\sigma = 10$.
- Number of extra loaves:

$$0.674 \times 10 \cong \textbf{6 loaves}$$

- **Optimal Stocking Quantity:**

$$\mathbf{Q^* = 90 + 6 = 96 \text{ loaves}}$$

Optimizing the Newsvendor Inventory Policy

Balance overage and underage costs to achieve the target service level.

01. The Economic Trade-off

Underage Cost (C_u): Profit lost per unit of unmet demand.
Overage Cost (C_o): Loss incurred for each unsold unit.

02. Critical Ratio (P)

The optimal service level is probability of not stocking out:

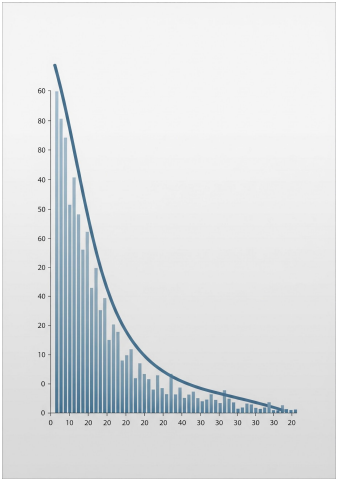
$$P = \frac{C_u}{C_o + C_u}$$

03. Stocking Decision

Convert target service level P into a z-value to determine optimal quantity $Q^* = \mu + z\sigma$.

Handling Non-Normal Demand Distributions

The Critical Ratio method remains valid even when demand isn't Gaussian.



01. Discrete Distributions

If demand is discrete (e.g., individual items), find the smallest quantity Q such that cumulative probability satisfies:

$$F(Q) \geq P \iff \text{Prob}\{D \leq Q\} \geq \text{Service Level}$$

02. Empirical Distributions

Use historical sales data directly to create a cumulative frequency table, then choose the quantity corresponding to target percentile P.

Uncertainty in Demand and Lead Times

Transitioning from fixed assumptions to probabilistic modeling.

Deterministic Models

Assume exact knowledge of demand and lead time. Everything is treated as a fixed constant.

Stochastic Models

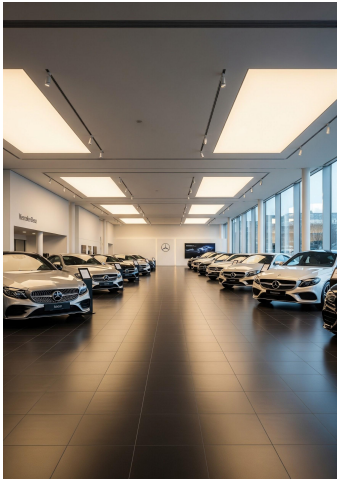
Demand and lead time become random variables. We manage the probability distribution of outcomes.

Primary Adaptation

Treat Lead Time Demand (LTD) as a combined random variable to determine safety stocks and reorder points.

Motivating Case: Car Dealership Inventory

Balancing luxury stock costs against the risk of lost high-value sales.



Scenario: Julia's Mercedes Dealership

- Monthly sales range from **6 to 12 cars**.
- Lead times vary between **1 to 4 months**.
- Current Policy: Order **14 cars** when stock hits **12**.
- **Goal:** Determine optimal order quantity (Q) and reorder point (r) to minimize total costs.

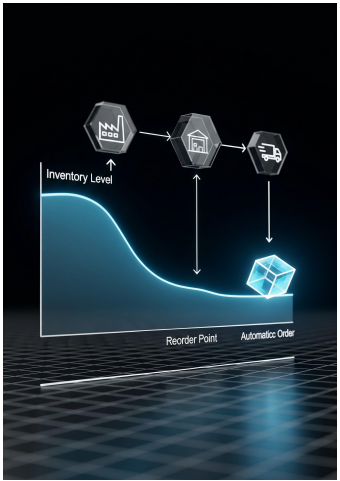
The High Stakes of Uncertainty

- Carrying Cost: **$600** / **car** / **month**.
- Ordering Cost: **$570** / **order**.

Stochastic modeling is required because both demand and delivery times are uncertain.

The (Q, r) Continuous-Review Policy

Optimizing replenishment for random demand and lead time scenarios.



Policy Fundamentals

- Monitor inventory position continuously in real time.
- Order lot size Q whenever position reaches reorder point r.
- Addresses simultaneous random demand and random lead time.
- Balances ordering costs, holding costs, and stockout risk.

Key Performance Measures

Average Inventory: $\approx \frac{Q}{2} + \text{Safety Stock}$

Cycle Service Level: Probability of no stockout occurring during a single replenishment cycle.

Order Quantity Q: The EOQ Foundation

Applying the Economic Order Quantity formula in stochastic environments.

The Classic EOQ Formula

$$\mathbf{Q} = \sqrt{\frac{\mathbf{2DS}}{\mathbf{H}}}$$

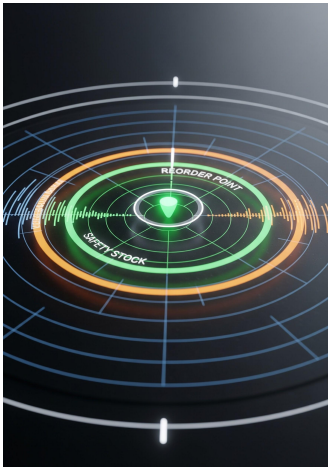
D: Avg Annual Demand S: Fixed Setup Cost
H: Holding Cost per Unit/Year

Why It Still Works

- **Cost Optimization:** Minimizes sum of setup and holding costs.
- **Stochastic Validity:** Valid renewal approximation under random demand.
- **Decoupled Variables:** Lead-time variability does not alter Q; it is handled by r.
- **Robustness:** EOQ curve is flat near the minimum; minor estimation errors carry negligible penalties.

Reorder Point r: Managing Uncertainty

Defining the inventory level that triggers a replenishment order.



The Reorder Point Formula

$$\mathbf{r} = \mu_{\mathbf{DLT}} + \text{Safety Stock} \quad (\mu_{DLT} = \bar{d} \times \bar{L})$$

$\bar{d}$: Avg demand per unit time $\quad \bar{L}$: Avg lead time (same units)

Safety Stock Calculation

$$\text{Safety Stock} = \mathbf{z} \cdot \sigma_{\mathbf{DLT}}$$

- z: Safety factor (from normal distribution table)
- σ_{DLT}: Standard deviation of lead-time demand
- *Example:* $z = 1.65$ provides a 95% cycle service level.

Safety Stock & Lead-Time Demand Variance

Calculating uncertainty when both demand and lead time are stochastic.

The Combined Variance Formula

$$\sigma_{DLT} = \sqrt{\bar{L}\sigma_d^2 + (\bar{d}\sigma_L)^2}$$

- σ_d^2: Variance of demand per unit time
- σ_L^2: Variance of lead time

Special Cases & Full Reorder Point

- Constant Lead Time ($\sigma_L = 0$): $\sigma_{DLT} = \sigma_d\sqrt{\bar{L}}$
- Constant Demand ($\sigma_d = 0$): $\sigma_{DLT} = \bar{d}\sigma_L$
- **Full Reorder Point (r):**

$$r = \bar{d}\bar{L} + z\sqrt{\bar{L}\sigma_d^2 + (\bar{d}\sigma_L)^2}$$

Case Study: Solving the Dealership Challenge

Applying combined variance formulas to Julia's Mercedes parameters.

Step 1: Identify Input Parameters

- Demand: $\bar{d} = 9$ cars/mo; $\sigma_d = 1.73$ (from 6–12 range)
- Lead Time: $\bar{L} = 2.5$ mos; $\sigma_L = 0.87$ (from 1–4 range)
- Service Level: 95% $\rightarrow z = 1.65$

Step 2: Calculate Combined Variance (σ_{DLT})

$$\sigma_{DLT} = \sqrt{2.5(3) + (9 \cdot 0.87)^2} \approx \textbf{8.3 cars}$$

Step 3: Determine Reorder Point (r)

$$r = \bar{d}\bar{L} + z \cdot \sigma_{DLT} = (9 \cdot 2.5) + (1.65 \cdot 8.3) \approx \textbf{36 cars}$$

$$\text{Safety Stock} = 1.65 \cdot 8.3 = \textbf{13.7 cars}$$

Case Study: Solving the Order Quantity

Determining the optimal number of Mercedes to order for Julia's dealership.

EOQ Recalculation: $Q \approx 4.1$ cars

Using $Q = \sqrt{\frac{2DS}{H}}$ with $D = 108$/yr, $S = \$570$, $H = \$7{,}200$/yr, optimal order quantity is $\approx$ **4 cars**, not 14!

Why 14 Cars is Inefficient

- **Excessive Holding Cost:** Creates average inventory of 7 cars ($\sim$ \$4,200/mo in holding costs).
- **Imbalance:** Over-prioritizes order frequency savings (\$570) at the expense of tied-up capital (\$600/car/mo).
- **Capital Drag:** Lot size of 4 aligns inventory flow with monthly demand of 9 cars.

Conclusion: The current 14-car policy is sub-optimal and significantly increases total cost.